

DFC ERI - Costing Model & Evidence Base

Draft v0.1 - Finance pillar working model Companion to 00 - Strategic Proposal

⚠ Read this before quoting any number

A deep evidence search (27 sources fetched, 104 claims extracted, 25 adversarially verified, 19 confirmed) found that **Nigeria has no reliable, current published unit-cost dataset for EMS** - and several specific figures circulating online were *refuted* under 3-vote verification (see [Evidence Base](#)). Therefore **every unit cost below is a transparent bottom-up planning assumption**, marked [ASSUMPTION – VALIDATE], anchored where possible to a verified data point or an international benchmark. These are *starting line items for the Finance pillar to challenge with real quotes*, not quoted prices. Do not submit these numbers to government as firm without validation.

FX assumption: ₦1,550 = US\$1 (mid-2026). [ASSUMPTION – VALIDATE]

1. Unit-cost assumptions (the inputs)

#	Line item	Planning unit cost	Basis / anchor	Confidence
U1	BLS ambulance - locally converted van (HiAce/Transit class), fully equipped	₦45 m capex [ASSUMPTION – VALIDATE]	No verified NG price; international BLS equipped ≈ \$25-40k, uplifted for NG import/conversion + 2025 devaluation	Low
U2	ALS ambulance - purpose-built, advanced equipment	₦110 m capex [ASSUMPTION – VALIDATE]	International ALS ≈ \$50-80k equipped, uplifted	Low
U3	Ambulance operating cost - crew + fuel + maintenance + comms	₦2.2 m / month (₦26.4 m/yr) [ASSUMPTION – VALIDATE]	Benchmark: GVK Dial 108 ≈ \$1.5-2.5k/ambulance/month	Low
U4	EMT training - per trainee, 3-6 month course	₦450 k [ASSUMPTION – VALIDATE]	NG paramedic/EMT course blogs (unverified); accreditation route TBC	Low

#	Line item	Planning unit cost	Basis / anchor	Confidence
U5	Paramedic up-skill - per trainee	₦1.5 m [ASSUMPTION – VALIDATE]	Higher-tier course estimate	Low
U6	Community first responder - training + bleeding-control kit, per head	₦40 k [ASSUMPTION – VALIDATE]	Anchored to ERA Edo pilot ≈ \$147 (~₦228k) per responder deployed with ambulance linkage - we separate training (~₦40k) from deployment/equipment	Medium
U7	Dispatch seat - setup (workstation, CAD terminal, telephony, headset)	₦3 m / seat [ASSUMPTION – VALIDATE]	Engineering estimate	Low
U8	Dispatch seat - opex (operator salary + telephony + facility share)	₦7 m / seat / yr [ASSUMPTION – VALIDATE]	Engineering estimate	Low
U9	Operator training - per operator	₦300 k [ASSUMPTION – VALIDATE]	Engineering estimate	Low
U10	CAD / dispatch software	Lean config ₦40-80 m one-time + ₦6-14 m/yr support [ASSUMPTION – VALIDATE]	Off-the-shelf SaaS \$15-50/seat/mo or DFC builds lean on existing stack (cheaper)	Medium
U11	Hospital trauma-bay upgrade - per hospital (monitors, defib, airway, suction, trauma trolley, FAST ultrasound, supplies)	₦30 m [ASSUMPTION – VALIDATE]	Engineering estimate; ₦15m basic → ₦100m full	Low
U12	Highway stabilisation post (Corridor tier only)	₦40 m capex + ₦18 m/yr opex [ASSUMPTION – VALIDATE]	Engineering estimate	Low
U13	Trauma fund - average per funded case (first-48hr stabilisation)	₦250 k [ASSUMPTION – VALIDATE]	UCH Ibadan ₦42,946 (2015) <i>refuted as firm anchor</i> but directional lower-bound; uplifted for severity mix + inflation + devaluation. Range ₦150k-₦600k	Low
U14	Operator onboarding & credentialing - per	₦400 k [ASSUMPTION – VALIDATE]	Engineering estimate; one-time	Low

#	Line item	Planning unit cost	Basis / anchor	Confidence
	aggregated vehicle (inspection, certification, platform onboarding)	VALIDATE]		
U15	Equipment top-up grant - per aggregated vehicle, partial subsidy to meet BLS standard	₦2 m [ASSUMPTION – VALIDATE]	Engineering estimate; far cheaper than buying a vehicle (U1)	Low
U16	Availability retainer - per priority on-call vehicle / month	₦250 k [ASSUMPTION – VALIDATE]	Engineering estimate; keeps a subset guaranteed-available	Low

Cross-cutting loadings applied to every tier: PMO / M&E / governance **+15%** of direct cost; **+10%** contingency; awareness campaign + community survey as a separate line.

Per-dispatch payments are not DFC capex. In the asset-light model (§2A) the payment for each actual trip flows from the funding stack (trauma fund / BHCPF EMT Gateway / insurer) *through* DFC's clearinghouse to the operator. It is a pass-through, not a DFC balance-sheet cost. We model only a modest Year-1 **bridge/float** where reimbursement lags.

2. Tier build-up (capex + Year-1 opex, excluding trauma fund)

2.1 LEAN - 1 LGA each in Lagos + FCT (2 sites)

Item	Qty	Unit	Capex	Yr-1 opex
BLS ambulances	4	U1/U3	₦180.0 m	₦105.6 m
EMTs trained	40	U4	₦18.0 m	-
Paramedics	8	U5	₦12.0 m	-
Community responders	300	U6	₦12.0 m	-
Dispatch / coordination cell	4 seats	U7/U8/U9	₦13.2 m	₦28.0 m
CAD software (lean)	1	U10	₦40.0 m	₦6.0 m
Hospital trauma bays	4	U11	₦120.0 m	-
Subtotals			₦395.2 m	₦139.6 m
Direct total (Yr-1)				₦534.8 m

Item	Qty	Unit	Capex	Yr-1 opex
PMO / M&E / governance (+15%)				₦80.2 m
Awareness + community survey				₦25.0 m
Contingency (+10%)				₦64.0 m
LEAN TOTAL (Yr-1)				≈ ₦704 m (~US\$454 k)

2.2 STANDARD - 2-3 LGAs per city (~5 LGAs)

Item	Qty	Unit	Capex	Yr-1 opex
Ambulances (10 BLS + 2 ALS)	12	U1/U2/U3	₦670.0 m	₦316.8 m
EMTs trained	120	U4	₦54.0 m	-
Paramedics	24	U5	₦36.0 m	-
Community responders	1,000	U6	₦40.0 m	-
Dispatch centre	8 seats	U7/U8/U9	₦26.4 m	₦56.0 m
CAD software	1	U10	₦60.0 m	₦10.0 m
Hospital trauma bays	10	U11	₦300.0 m	-
Subtotals			₦1,186.4 m	₦382.8 m
Direct total (Yr-1)				₦1,569.2 m
PMO / M&E / governance (+15%)				₦235.4 m
Awareness + community survey				₦50.0 m
Contingency (+10%)				₦185.0 m
STANDARD TOTAL (Yr-1)				≈ ₦2.04 bn (~US\$1.32 m)

2.3 CORRIDOR - cities + East-West road

Item	Qty	Unit	Capex	Yr-1 opex
Ambulances (18 BLS + 6 ALS)	24	U1/U2/U3	₦1,470.0 m	₦633.6 m
EMTs trained	250	U4	₦112.5 m	-
Paramedics	50	U5	₦75.0 m	-
Community responders	2,500	U6	₦100.0 m	-
Dispatch centre	12 seats	U7/U8/U9	₦39.6 m	₦84.0 m
CAD software	1	U10	₦80.0 m	₦14.0 m

Item	Qty	Unit	Capex	Yr-1 opex
Hospital trauma bays	18	U11	₦ 540.0 m	-
Highway stabilisation posts	4	U12	₦ 160.0 m	₦ 72.0 m
Subtotals			₦2,577.1 m	₦803.6 m
Direct total (Yr-1)				₦3,380.7 m
PMO / M&E / governance (+15%)				₦ 507.0 m
Awareness + community survey				₦ 90.0 m
Contingency (+10%)				₦ 398.0 m
CORRIDOR TOTAL (Yr-1)				≈ ₦4.38 bn (~US\$2.82 m)

2A. Asset-light aggregation variant (recommended for the ambulance line)

Section 2 assumes DFC **buys** the ambulance fleet. The strategy (Proposal §4) is the opposite: DFC **aggregates existing private capacity**. This variant replaces the ambulance *capex* (U1/U2) and the operator-borne *opex* (U3) with a much smaller onboarding + equipment-top-up + retainer cost (U14/U15/U16), because the operators already own and run the vehicles and recover their per-trip cost through the funding stack. Everything else (training, dispatch, CAD, hospital bays, loadings) is unchanged.

The logic: instead of ~~₦~~45 m to buy and ~~₦~~26.4 m/yr to run each ambulance, DFC pays ~~~₦~~400 k to credential an existing one, optionally ~~~₦~~2 m to bring it to standard, and ~~~₦~~250 k/month to retain a priority subset on-call. The per-trip cost is reimbursed, not capitalised.

2A.1 LEAN - asset-light

Item	Qty	Unit	Capex	Yr-1 opex
Aggregated vehicles - onboarding	25	U14	₦ 10.0 m	-
Equipment top-up grants	25	U15	₦ 50.0 m	-
Availability retainer (priority subset)	10 veh	U16	-	₦ 30.0 m
Year-1 reimbursement bridge/float	-	-	-	₦ 20.0 m
EMTs / paramedics / responders	40/8/300	U4/U5/U6	₦ 42.0 m	-
Dispatch / coordination cell	4 seats	U7/U8/U9	₦ 13.2 m	₦ 28.0 m
CAD software (lean)	1	U10	₦ 40.0 m	₦ 6.0 m

Item	Qty	Unit	Capex	Yr-1 opex
Hospital trauma bays	4	U11	₦120.0 m	-
Subtotals			₦275.2 m	₦84.0 m
Direct total (Yr-1)				₦359.2 m
PMO / M&E / governance (+15%)				₦53.9 m
Awareness + community survey				₦25.0 m
Contingency (+10%)				₦35.9 m
LEAN TOTAL - asset-light (Yr-1)				≈ ₦474 m (~US\$306 k)

2A.2 STANDARD - asset-light

Item	Qty	Unit	Capex	Yr-1 opex
Aggregated vehicles - onboarding	60	U14	₦24.0 m	-
Equipment top-up grants	60	U15	₦120.0 m	-
Availability retainer	25 veh	U16	-	₦75.0 m
Year-1 reimbursement bridge/float	-	-	-	₦40.0 m
EMTs / paramedics / responders	120/24/1,000	U4/U5/U6	₦130.0 m	-
Dispatch centre	8 seats	U7/U8/U9	₦26.4 m	₦56.0 m
CAD software	1	U10	₦60.0 m	₦10.0 m
Hospital trauma bays	10	U11	₦300.0 m	-
Subtotals			₦660.4 m	₦181.0 m
Direct total (Yr-1)				₦841.4 m
PMO / M&E / governance (+15%)				₦126.2 m
Awareness + community survey				₦50.0 m
Contingency (+10%)				₦84.1 m
STANDARD TOTAL - asset-light (Yr-1)				≈ ₦1.10 bn (~US\$711 k)

2A.3 CORRIDOR - asset-light (hybrid)

The remote East-West highway stretches have little idle private capacity to aggregate, so this tier stays **hybrid**: aggregate in the cities, but buy a small dedicated highway ALS fleet + posts.

Item	Qty	Unit	Capex	Yr-1 opex
City vehicles - onboarding	90	U14	₦36.0 m	-
Equipment top-up grants	90	U15	₦180.0 m	-
Dedicated highway ALS (owned)	6	U2/U3	₦660.0 m	₦158.4 m
Highway stabilisation posts	4	U12	₦160.0 m	₦72.0 m
City availability retainer	40 veh	U16	-	₦120.0 m
Year-1 reimbursement bridge/float	-	-	-	₦60.0 m
EMTs / paramedics / responders	250/50/2,500	U4/U5/U6	₦287.5 m	-
Dispatch centre	12 seats	U7/U8/U9	₦39.6 m	₦84.0 m
CAD software	1	U10	₦80.0 m	₦14.0 m
Hospital trauma bays	18	U11	₦540.0 m	-
Subtotals			₦2,022.6 m	₦508.4 m
Direct total (Yr-1)				₦2,531.0 m
PMO / M&E / governance (+15%)				₦379.7 m
Awareness + community survey				₦90.0 m
Contingency (+10%)				₦253.1 m
CORRIDOR TOTAL - asset-light (Yr-1)				≈ ₦3.25 bn (~US\$2.10 m)

2A.4 Owned-fleet vs asset-light - the headline comparison

Tier	Owned fleet (\$2)	Asset-light aggregation (\$2A)	Year-1 saving
Lean	₦0.70 bn	₦0.47 bn	~33%
Standard	₦2.04 bn	₦1.10 bn	~46%
Corridor (hybrid)	₦4.38 bn	₦3.25 bn	~26%

Why asset-light is the recommended basis: it is cheaper, it matches the actual strategy (Proposal §4), it raises utilisation of vehicles that already exist, and it makes operators paying participants in a sustainable system rather than capital DFC has to raise and depreciate. The owned-fleet figures (\$2) are retained only as the upper bound / fallback if aggregation onboarding proves slower than planned.

3. Trauma fund (Tier 2 add-on) - sized separately

Per the strategy, the fund is **not** in the core ask; it is the recurring liability that statutory funding (BHCPF EMT Gateway / NHIA) is meant to assume. Annual sizing = funded RTA-trauma cases × U13 (₹250k/case).

Tier	Modelled funded cases/yr [ASSUMPTION – VALIDATE vs local FRSC + hospital data]	Annual trauma-fund cost
Lean	~1,500	≈ ₹375 m / yr
Standard	~4,000	≈ ₹1.0 bn / yr
Corridor	~8,000 (high highway-trauma volume)	≈ ₹2.0 bn / yr

National context for sizing: FRSC recorded **5,421 road deaths in 2024** (verified); injured volumes are several multiples of deaths. Pilot case volumes must be validated against **LGA-level FRSC crash data + pilot-hospital trauma registers** before the fund is committed.

4. Year-2+ recurring cost (sustainability view)

Capex is largely Year-1. Steady-state annual run cost ≈ **opex + loadings** (no re-purchase of ambulances/equipment, modest replacement reserve):

Tier	Approx. annual recurring (ex-trauma-fund)	+ Trauma fund	Steady-state total/yr
Lean	≈ ₹200 m	₹375 m	≈ ₹575 m/yr
Standard	≈ ₹520 m	₹1.0 bn	≈ ₹1.52 bn/yr
Corridor	≈ ₹1.05 bn	₹2.0 bn	≈ ₹3.05 bn/yr

This is the number the funding stack must cover in perpetuity - and why grants (which fund the Year-1 build) cannot be the sustainability answer. The statutory + insurance + micro-levy layers carry the recurring cost. See proposal §3.

5. International benchmarks (for credibility, not as NG prices)

Model	What it tells us	Source quality
India - GVK EMRI "Dial 108"	The reference PPP: government largely funds operating cost, private operator runs the system; ~1 ambulance per ~100k population as a coverage target; per-trip subsidy economics that scale down with	Secondary (Wikipedia + study)

Model	What it tells us	Source quality
	volume. Validates the <i>DFC-operates / government-funds-per-case</i> structure.	
ERA (Emergency Response Africa), Edo, Nigeria	A Nigerian community-first-responder + ambulance pilot; ~167 CBFRs trained; cost anchor $\approx$ \$147/responder-episode . Closest real local data point we have.	Primary (funder report)
Sub-Saharan Africa GDP loss	SSA countries lose $\geq$ 0.8% of GDP to road traffic injuries - the macro case for investment.	Verified

Refuted / do NOT cite as firm (failed 3-vote verification): UCH Ibadan "~~N~~42,946 per RTI victim"; "8% vs 40%" prehospital-vs-no-care mortality; a sub-Saharan "INT\$ per injury" range; the "33% transport / 30.5% financial barrier" split; and a specific 3-gateway/NEMTC description of BHCPF. Use only as directional context with the caveat attached.

6. Cost-effectiveness framing (what the pilot must *prove*, not claim)

We deliberately do **not** assert a mortality-reduction figure - the headline stat circulating online failed verification. Instead, the pilot's core evidence output is its own cost-per-life-saved, measured by independent M&E against the FRSC baseline. The illustrative arithmetic - Lean tier at $\approx$ ~~N~~700m Year-1 against thousands of preventable trauma deaths nationally - is favourable enough that the pilot only has to demonstrate a *modest* golden-hour effect to be among the most cost-effective health interventions available. That measured number becomes the single most powerful asset for the scale-up funding case.

Evidence Base

Verified anchors used (survived 3-vote adversarial verification): - BHCPF established under **National Health Act 2014 §11**, funded by an earmark of $\geq$ **1% of the Consolidated Revenue Fund**. - **EMT Gateway Fund** operates as a window of the BHCPF and funds NEMSAS free-48-hour emergency care. - **NEMSAS** mandate guarantees emergency medical treatment; reports **5,255 beneficiaries in 6 months**; expanded to 30 states. - **Free 48-hour emergency treatment policy** launched (government). - **BHCPF can accept international donor contributions** - relevant to blending grant + statutory funding. - All Nigerian **telecom operators mandated to route 112** - relevant to the comms layer + MTN partnership. - **FRSC: 5,421 road deaths in 2024**. - **86% of road-crash victims faced catastrophic health expenditure** (Nigerian study). - **SSA loses $\geq$ 0.8% of GDP** to road traffic injuries. - ERA community-responder pilot: ~167 CBFRs;

≈\$147 cost anchor. - "Few high-quality EMS cost-effectiveness studies exist" for this context - itself the reason this model is bottom-up.

Key sources (full list of 27 in the research transcript): - NEMSAS - <https://nemsas.com/> - NHIA / BHCPF - <https://www.nhia.gov.ng/basic-health-care-provision-fund/> - World Bank, BHCPF tariff & payment options - <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/465201615805182969/> - Guardian - NEMSAS EMT Gateway beneficiaries - <https://guardian.ng/news/5255-nigerians-benefited-from-emt-gateway-fund-in-6-months-nemsas/> - Gazette NGR - NEMSAS expands to 30 states - <https://gazettengr.com/nemsas-expands-emergency-medical-coverage-to-30-states/> - TV360 - free 48-hour policy - <https://www.tv360nigeria.com/nigeria-government-launches-free-48-hour-emergency-treatment-policy/> - Punch / FRSC - 5,421 road deaths 2024 - <https://punchng.com/road-crashes-claimed-5421-lives-in-2024-frsc/> - NCC - Emergency Communications Centres (112) - <https://ncc.gov.ng/the-ncc/stakeholders/projects/74-emergency-communications-centres-ecc-ERA-community-first-responder-pilot> - <https://fundinnovation.dev/en/projects/improving-emergency-response-with-a-network-of-trained-community-based-first-responders-in-nigeria> - GVK EMRI (Dial 108) - https://en.wikipedia.org/wiki/GVK_EMRI - PSHAN Vivo Health - <https://businessday.ng/health/article/pshan-launches-vivo-health-app-to-fix-health-emergency-response-gaps-in-nigeria/> - JEMS - improving care response in Nigeria - <https://www.jems.com/ems-operations/ground-ambulance-operations/improving-care-response-in-nigeria/>

Caveats (verbatim from the research synthesis): *No Nigeria 2024-26 unit costs; comparables unconfirmed; FRSC incidence (injured, by LGA) missing.* The Finance pillar must source real quotes for U1-U12 and local crash/trauma volumes for the trauma-fund sizing before submission.

Evidence Base - Ambulance Aggregation (Section 4 of the Proposal)

A second adversarially-verified research pass (111 agents; 24 claims confirmed 3-0, 1 magnitude claim refuted) produced these anchors for the aggregation model:

Flare / Rescue.co (Kenya): - Owns no ambulances; aggregates existing private providers via formal SLAs; provides dispatch tech, GPS, triage training, hospital-referral dashboards; assesses operators for service/personnel/equipment quality. - **800+ public and private providers** aggregated by 2025; **47,000+ responses across all 47 counties; 2,000 health facilities** linked. - Subscription **~KES 4,000 (~€33)/subscriber/yr**; insurance risk-pooling integration as a reimbursement channel. - Source: [RSM](#), [PMC PMC12418930](#), [Oxford OODH](#).

India 108 / GVK EMRI: - Single-operator PPP: **9,655 ambulances, 66,472+ employees, 17 states + 1 UT, ~1.02 bn covered.** - Governed by **one MOU per state**; values e.g. Rajasthan ₹50 cr/yr (capex+opex), AP ₹114 cr/yr opex, Gujarat ₹252 cr/5 yrs. - **Government funds ~95% of**

opex (100% capital in Gujarat/Rajasthan); operator contribution falls 25%→5% by year 2; free + toll-free to patients. - Source: [EMRI](#), [NHSRC India](#).

ERA (Edo State, Nigeria): - Govt EMS partnership (Edo govt, UBTH, Edo State Health Insurance Scheme); pilot from **5 Mar 2024, 15 months**. - **170 CBFs across 3 LGAs** (Oredo, Egor, Ikpoba-Okha); **4 ambulances** at PHCs; 10-min access goal; 24/7 Command Center; dispatch linked to **112 + 739**; three apps. - Source: [ERA Edo EMS](#), [ERA About](#).

Credentialing standard: - **CAAS Ground Vehicle Standard** - minimum vehicle requirements + certify-then-mark compliance symbol; non-compliant excluded. Source: [CAAS](#).

Structural / financing facts: - Decentralised systems with no coordination cap coverage at ~71.5% regardless of profitability. Source: [Springer](#). - **~95% of Africa lacks a centralised EMS system**. Source: RSM (above). - **BHCPF: ~5% via EMT Gateway to NEMSAS reimburses accredited private AND public ambulance providers** - the statutory rail for the aggregation model. (*Confirm 5%/NEMTC governance detail at source with NEMSAS.*)

Refuted (do not cite as fact): the specific "Nairobi 162 min → under 15 min" response-time magnitude (one framing confirmed, a near-duplicate refuted 1-2) - present only as operator-reported.